Padraig Lavery

Full Stack Software Engineer – Building with LLMs, RAG & Agents

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Full stack software engineer with 4+ years building and running production systems for UK and Middle East banking clients (Angular, .NET/C#, Node.js, PostgreSQL on AWS and Kubernetes). Building applied AI skills through Incident Copilot, a triage service using retrieval-augmented generation and agentic tool-calling, and preparing for the AWS Certified AI Practitioner exam. Takes ownership end to end: leads projects, mentors new starters and leaves systems easier to run.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Full stack ownership

- Built full stack features across web and mobile, including a secure e-code resend flow with audit-logged resends and multi-bank email routing, and targeted customer redemption pop-ups with deep-linking; fixed WCAG colour-contrast failures (1.7:1 to 4.5:1).
- Led the technical migration of email templates onto dedicated brand domains across four banking brands (MJML, .NET Core).
- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals.
- Built the CI/CD pipeline for a UI Builder service as part of an internal AI investigation project.

Production and quality

- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation.
- Eliminated recurring daily out-of-memory crashes in a high-volume ingestion service by loading caches once per batch instead of per file.
- Led a cross-team initiative for automated API testing: request validation, shared error handling, OpenAPI documentation, unit tests and stabilised GitHub Actions CI.
- Hardened authentication and account services and remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance.

Team

- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost web API using Go, Docker, Kubernetes and Git.

AI PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Takes a production incident description, retrieves relevant runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status.
- Moved from AWS Bedrock to locally hosted models via Ollama to remove inference costs. Runbook knowledge base and FastAPI service in place; retrieval, model integration and testing in development.

ML Trading System (in progress) | Python, Alpaca API

- Built a backtester for SEC Form 4 insider purchases by company CEOs, then extended it into an ML-driven strategy running on a paper-trading account.

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular card transactions, with a model trained in R and Python and served via SageMaker.

SKILLS

- **AI:** LLM integration, retrieval-augmented generation, agentic tool-calling, pgvector, Ollama, AWS Bedrock, SageMaker
- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL

- **Front end:** Angular, HTML/CSS, WCAG accessibility
- **Back end and data:** Node.js, .NET Core, FastAPI, Flask, GraphQL, PostgreSQL, MongoDB
- **Cloud and delivery:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Docker, Terraform, GitHub Actions

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026